

Vijay Saravanan

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SUMMARY

Backend Engineer with expertise in Go, C++, and distributed systems. M.Tech CSE at IIT Kharagpur with research in graph algorithms and computational complexity. Ex-ColorTokens SDET who automated 50+ API test suites in Golang, cut manual QA effort by 40%, and protected Fortune-class clients on the XShield cybersecurity SaaS platform. Built a Redis-class in-memory key-value store from scratch achieving 58K ops/sec. Seeking SDE / Backend / Platform Engineering roles at product-first companies.

EDUCATION

Indian Institute of Technology, Kharagpur

M.Tech, Computer Science and Engineering | CGPA: 8.22/10

2024 – 2026

Kharagpur, West Bengal

College of Engineering, Guindy – Anna University

B.Tech, Information Technology | CGPA: 7.78/10

2019 – 2023

Chennai, Tamil Nadu

PROFESSIONAL EXPERIENCE

ColorTokens Inc.

July 2023 – March 2024

Member of Technical Staff 1 – SDET

Bengaluru, India

- Contributed to **XShield**, a SaaS cybersecurity platform for configuring micro-segmentation policies that prevent lateral malware propagation across enterprise servers, routers, and IoT devices for Fortune-class clients.
- Designed and automated **50+ API test suites in Golang** across microservices, reducing manual QA effort by **40%** and improving release confidence.
- Built and managed daily regression testing pipelines with auto-generated stakeholder reports, accelerating release validation and deployment cycles.
- Identified and resolved critical Node.js vulnerabilities and stability bugs, directly improving uptime and deployment reliability in production.

ACADEMIC PROJECTS (IIT KHARAGPUR)

Equitable Graph Coloring for Structured Graph Classes

Jun 2025 – Apr 2026

C++, Python, Graph Theory, Algorithm Design

- M.Tech thesis research: developed polynomial-time algorithms for the equitable coloring problem on circular-arc graphs, block graphs, and bipartite outerplanar graphs.
- Analysed computational complexity bounds and constructed formal proofs deriving tight chromatic-number bounds with applications to scheduling and load balancing.
- Skills:** Algorithm Design, Graph Theory, Computational Complexity, Theoretical Computer Science, Mathematical Proof, Research.

BLINK DB – High-Performance In-Memory Key-Value Store

Jan 2025 – Mar 2025

C++, Linux, epoll, Bash, Redis

- Built a Redis-class in-memory key-value database using **epoll-based I/O multiplexing**, handling **10,000+ concurrent client connections** on a single thread.
- Achieved **58K ops/sec (SET)** and **47K ops/sec (GET)** under benchmarking, validating Redis-level performance characteristics.
- Implemented custom eviction and persistence mechanisms to optimise memory usage and scalability under sustained load.
- Skills:** System Programming, Database Internals, Concurrency, Epoll, Performance Engineering, C++, Linux.

Air Quality Monitoring System – Real-Time Data Platform

Oct 2024 – Nov 2024

React.js, Docker, REST API

- Built a real-time AQI dashboard integrating the World Air Quality Index (WAQI) API to track live pollution data across PM2.5, PM10, and other key pollutants.
- Containerised the full application with **Docker** for consistent, scalable deployment across environments.
- Designed an interactive React.js dashboard with responsive components and API-polling logic for live refresh without page reloads.
- Skills:** Docker, React.js, REST API Integration, Real-Time Systems, Frontend Engineering, Containerisation.

BlockCart – Decentralised E-Commerce Platform on Ethereum

Aug 2024 – Sep 2024

Solidity, Ethereum, Web3, Smart Contracts

- Designed and deployed Solidity smart contracts handling the full product lifecycle: listing, purchase, and ownership transfer with cryptographic guarantees.
- Leveraged blockchain immutability to prevent data tampering and ensure transparent, trustless transactions without a central authority.
- Deployed and validated on the Ethereum test network through simulated multi-user transaction flows.
- Skills:** Blockchain, Solidity, Ethereum, Smart Contracts, Decentralised Systems, Web3.

UNDERGRADUATE PROJECTS (CEG, ANNA UNIVERSITY)

Face Super-Resolution using DCGAN and SRGAN

Aug 2022 – Apr 2023

Python, TensorFlow, Deep Learning, Flask

- Built a two-stage deep learning pipeline to reconstruct occluded and low-resolution face images using Generative Adversarial Networks for occlusion-robust face recognition.
- Stage 1: trained a **Deep Convolutional GAN (DCGAN)** to inpaint missing or occluded facial regions from surrounding visual context.
- Stage 2: applied a **Super-Resolution GAN (SRGAN)** to upscale and sharpen the reconstructed image, improving perceptual quality and recognition accuracy.
- Deployed the trained model as a **Flask REST API**, enabling real-time inference from web clients for surveillance and identity verification use cases.
- **Skills:** Deep Learning, GANs, Computer Vision, TensorFlow, Python, Flask, Model Deployment.

Flight Delay Prediction Model

Aug 2022 – Dec 2022

Python, Machine Learning, Flask

- Built a machine learning model to predict commercial flight delays using large-scale, multi-parameter aviation datasets covering weather, route, airline, and historical delay signals.
- Performed data cleaning, **feature engineering**, and class-imbalance handling on high-dimensional flight records to extract predictive signals.
- Trained and benchmarked multiple classification models, selecting the best-performing predictor for delay-likelihood estimation.
- Deployed the model as a **Flask web service** exposing REST endpoints for live delay-probability inference on user-supplied flight parameters.
- **Skills:** Machine Learning, Python, Data Engineering, Feature Engineering, Predictive Analytics, Flask, Model Deployment.

Hand Gesture Controlled Robot

Jan 2022 – May 2022

Arduino, C/C++, IoT, Embedded Systems

- Designed and built a gesture-controlled robotic vehicle that translates real-time hand orientation into directional motor commands, eliminating the need for a physical controller.
- Integrated **accelerometer sensors** on a wearable glove to capture 3D hand-tilt data and processed input on an **Arduino microcontroller**.
- Implemented wireless RF communication between glove and robot chassis, validated through field testing at an effective range of **10 metres**.
- Engineered the full embedded stack: hardware interfacing, sensor calibration, real-time signal processing, and wireless protocol design.
- **Skills:** Arduino, IoT, Embedded Systems, Sensor Integration, Real-Time Control, Hardware-Software Interface, C/C++.

TECHNICAL SKILLS

Languages (Proficient):	C++, Python, Go (Golang)
Languages (Experienced):	JavaScript (Node.js), Java, Shell Scripting, Solidity, SQL
Frameworks & Libraries:	React.js, Node.js, Flask, TensorFlow
Systems & Databases:	Distributed Systems, Redis, epoll, MapReduce, REST APIs, Microservices
DevOps & Cloud:	Docker, Kubernetes, Jenkins, Git, GitHub, Bitbucket, Azure DevOps
Tools & Platforms:	Postman, JIRA, Confluence, Selenium, Linux, Bash
Areas of Interest:	Backend Engineering, Cloud, SaaS, Operating Systems, Networks, Distributed Systems

POSITIONS OF RESPONSIBILITY

Teaching Assistant – PDS Lab & PDS Theory

Aug 2025 – Apr 2026

Indian Institute of Technology, Kharagpur

Kharagpur, India

- Mentored **50+ students** in Programming, Data Structures, and Algorithms through structured office hours, problem-solving sessions, and tutorial presentations.
- Evaluated and provided detailed feedback on 50+ class test papers and lab assignments, reinforcing core algorithmic principles.